Example 1

Blind Hole: Ø 15×45 mm on a stainless steel part

Data:

$$k_c = 2800 \text{ MPa}$$
$$\varepsilon = 140°$$
$$f = 0{,}2 \text{ mm/rev}$$
$$v_c = 60 \text{ m/min}$$

Determine:

a) Chip thickness
b) Force, torque and power
c) Drilling time $(e_{in} = 2 \text{ mm})$

Example 2

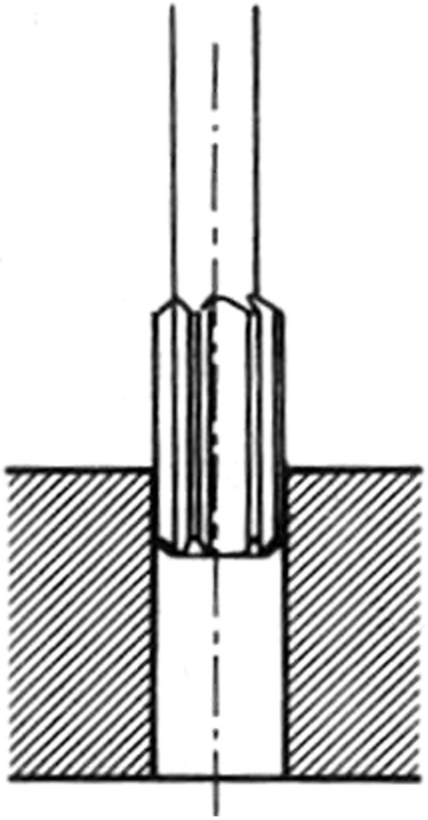
Through hole drilling and reaming (H7):
$\varnothing\ 20\ x\ 25\ \mathrm{mm}$ on carbon steel plate (HB200)

Drilling:

$$D = 19\ \mathrm{mm}$$
$$k_c = 2100\ \mathrm{Mpa}$$
$$\varepsilon = 140°$$
$$f = 0{,}1\ \mathrm{mm/rev}$$
$$v_c = 120\ \mathrm{m/min}$$

Reaming:

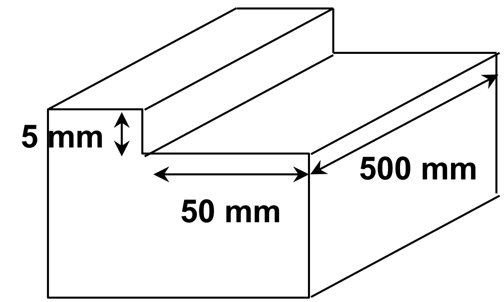
$$D = 20\ \mathrm{mm}$$
$$k_c = 2100\ \mathrm{Mpa}$$
$$Z = 20$$
$$f = 1{,}5\ \mathrm{mm/rev}$$
$$v_c = 180\ \mathrm{m/min}$$



Determine the cutting force, torque, and power for the two machining operations.

Example 3

The groove shown in figure should be made using a milling operation. Determine the corresponding cutting power.



Tool data:

- Number of tooth $Z = 5$;
- Entering angle $\kappa_r = 90^o$;
- Mill diameter $D = 60$ mm;

Machining parameters:

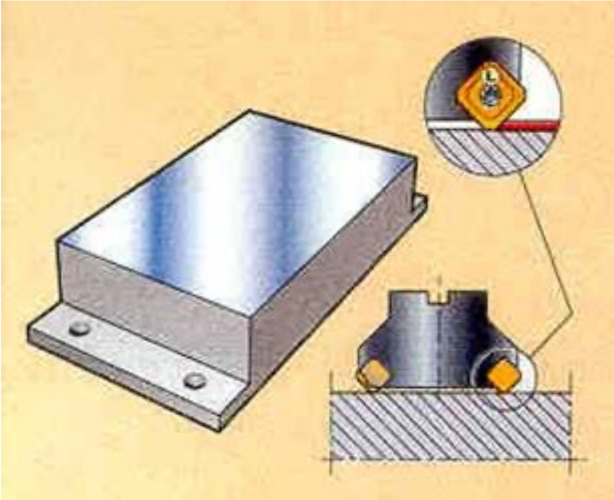
- Cutting pressure $k_{c0,4} = 1600$ MPa
- Coefficient x = 0,29
- Feed per tooth $f_z = 0{,}1$ mm/rev

Maximum chip thickness [mm]	0,05	0,1	0,2
Cutting speed [m/min]	305	275	225

Determine the cutting force, torque, power, and time for this machining operation.

Example 4

A cast iron plate, with hardness HB 120 and size 100 x 60 mm, needs to be machined by a milling operation. The amount of material allowance to be remove is equal to 8 mm.



Determine:

a) Chip thickness

b) Force, torque and power

c) Milling time

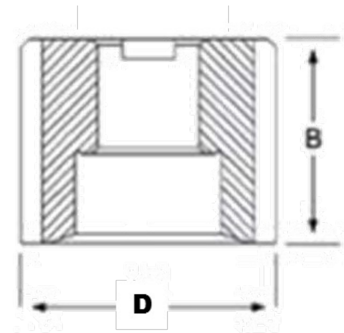
considering both peripheral milling and face milling.

Both situations should consider the following parameters k_{cs} = 900 MPa, x = 0,28, $e_{\mathrm{in}} = e_{out}$ = 5 mm, v_{c} = 100 m/min, and f_{z} = 0,3 mm/rev.

Example 4

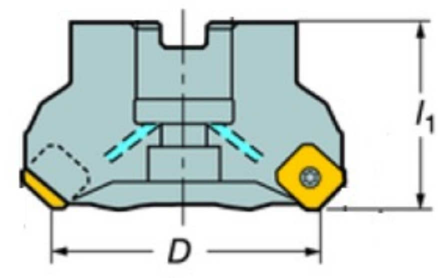
Slab milling tool:

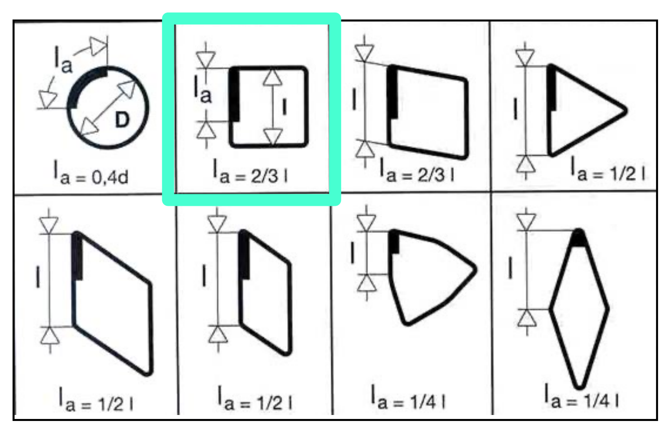
- B = 80 mm
- D = 32 mm
- Z = 8



Face milling tool:

- l_1 = 60 mm
- D = 100 mm
- Z = 12, κ_r = 45°
- Square inserts (l = 9 mm), κ_r = 45°
- D > L (1,3 L < D < 1,4 L)
- Entrance length ≈ 0,1 D
- Exit length ≈ 0,3 D





Example 5

A steel square planar surface with a length equal to 300 mm must be flattened. A roughing operation will remove a machining allowance equal to 4 mm. The face mill will be positioned initially so that 25% of the tool diameter will exit from one side of the plane width and the square plane is machined according to the path in the figure.

Determine:

1) The cutting torque and the cutting power, verifying that the operation can be carried out with a milling machine with a power of 15 kW and an efficiency of 80%.
2) The machining time needed to carry out the operation.

Data: mill diameter D = 80 mm; number of mill cutting edges Z = 8; main entering angle equal to 90°; Kronenberg parameters k_{cs} = 1600 MPa, x = 0,197; feed per tooth f_z = 0,15 mm/rev; cutting speed v_c = 250 m/min.

Example 5